

CBSE Class 10 Mathematics — Areas Related to Circles

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 24 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the circumference of a circle is 132 cm, then its radius is:
(A) (A) 14 cm
(B) (B) 21 cm
(C) (C) 28 cm
(D) (D) 42 cm
2. The area of a circle whose diameter is 14 cm is:
(A) (A) 44 cm^2
(B) (B) 77 cm^2
(C) (C) 154 cm^2
(D) (D) 308 cm^2
3. The area of a quadrant of a circle with radius $r = 7 \text{ cm}$ is:
(A) (A) 38.5 cm^2
(B) (B) 77 cm^2
(C) (C) 154 cm^2
(D) (D) 19.25 cm^2
4. If the perimeter of a semi-circular protractor is 36 cm, find its diameter.
(A) (A) 7 cm
(B) (B) 14 cm
(C) (C) 18 cm
(D) (D) 28 cm
5. The area of a sector of a circle with radius 6 cm and central angle 60° is:
(A) (A) $2\pi \text{ cm}^2$

- (B) (B) $3\pi \text{ cm}^2$
 (C) (C) $4\pi \text{ cm}^2$
 (D) (D) $6\pi \text{ cm}^2$
6. Assertion (A): The area of a circle with radius 7 cm is 154 cm^2 . Reason (R): The area of a circle is given by the formula πr^2 , where r is the radius.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The circumference of a circle with diameter 14 cm is 44 cm. Reason (R): The circumference of a circle is given by πd , where d is the diameter.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. Reason (R): A semicircle is half of a full circle.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The length of an arc of a circle with radius r and central angle θ is $\frac{\theta}{360} \times 2\pi r$. Reason (R): The circumference of a circle is $2\pi r$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): The area of a quadrant of a circle with radius r is $\frac{1}{4}\pi r^2$. Reason (R): A quadrant is a sector with a central angle of 45° .

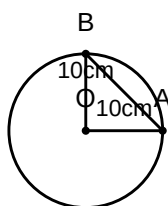
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

- Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° .
- The length of the minute hand of a clock is 14 cm . Find the area swept by the minute hand in 5 minutes.
- If the perimeter and the area of a circle are numerically equal, then find the radius of the circle.
- Find the area of a quadrant of a circle whose circumference is 44 cm .
- The radius of two circles are 8 cm and 6 cm respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

Section D: Long Answer (4-5 marks each)

- Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° . Use $\pi = 3.14$.
- In a circle of radius 10 cm , a chord AB subtends a right angle at the centre O . Find the area of the minor segment. (Use $\pi = 3.14$).



- The length of the minute hand of a clock is 14 cm . Find the area swept by the minute hand in 5 minutes.
- The diameters of two circles are 8 cm and 6 cm respectively. Find the radius of a circle whose area is equal to the sum of the areas of the two circles.

5. A horse is tethered to a peg at one corner of a square-shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze.
6. A round table cover of radius 28 cm has a design in the middle leaving an equilateral triangle ABC in the middle as shown in the figure. Find the area of the design (shaded region). Use $\sqrt{3} = 1.73$ and $\pi = \frac{22}{7}$.
7. A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115° . Find the total area cleaned at each sweep of the blades. (Take $\pi = \frac{22}{7}$)
8. In a circular park of radius 21 m , a path of uniform width 3.5 m is constructed inside the park along the boundary. Find the cost of paving the path at the rate of $\text{Rs.}50$ per m^2 .
9. A chord AB of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)